



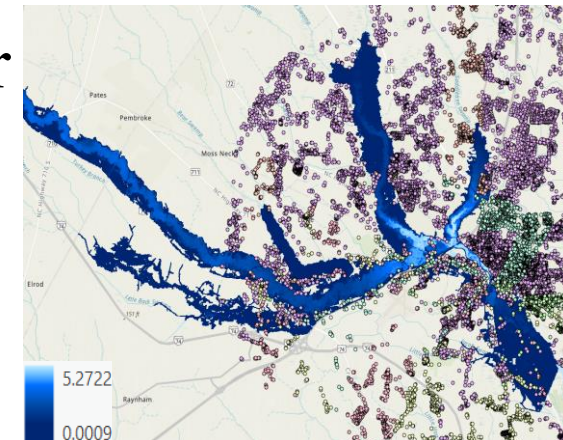
FloodGuard Miami

Predict. Prepare. Protect.

Smart, no-code flood risk platform for Miami's communities

Presented by:

Mostafa Sameer — Civil Engineering Innovator



The Problem



- Miami is one of the **most flood-prone cities in the U.S.**
 - Even *sunny-day flooding* affects neighborhoods like Brickell and Little River.
 - Existing FEMA maps are **outdated, reactive, and not localized.**
- City planners, residents, and businesses **lack real-time flood risk data.**



- Sea level projected to rise **14–26 inches by 2060.**
- **Over \$3B allocated** to flood resilience in Miami-Dade 2030 Plan.
- Demand for **predictive, affordable flood tools** is growing across South Florida.

Miami Mayor: City Flooding “Like a Hurricane” Again Today Thanks to King Tides

Thanks to sea-level rise, Florida's unique topography, and poor city planning, areas of Miami-Dade County look like a hurricane hit them today. But there's not even a tropical storm in town. Instead, mere weeks after a real hurricane did damage major parts of South Florida, the Miami area is massively flooding thanks to a combination of some moderate storms hitting during king tides, when the sea is at its highest point all year.

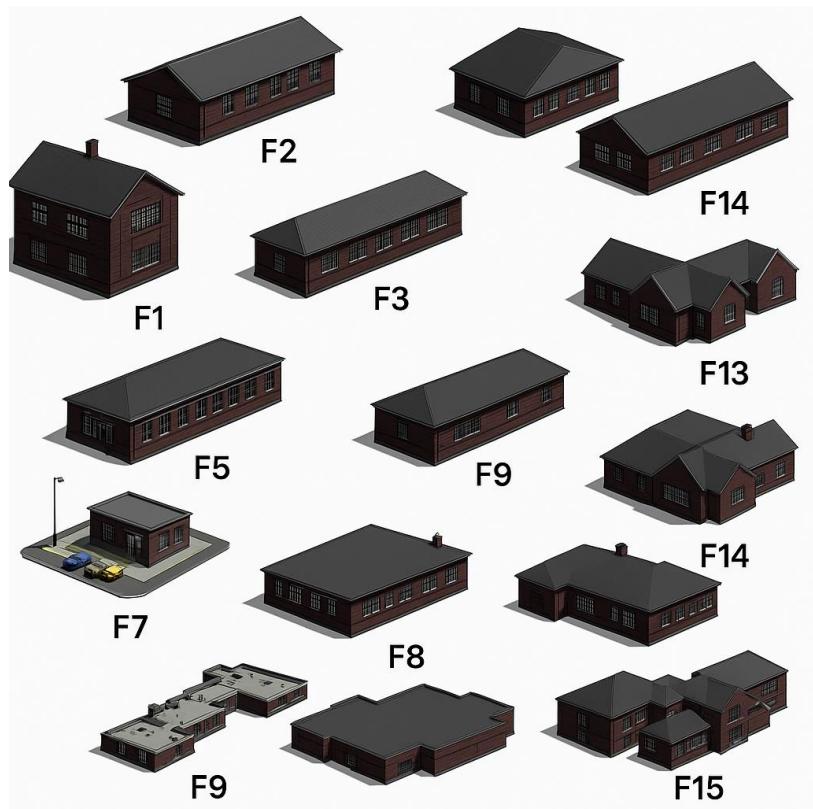
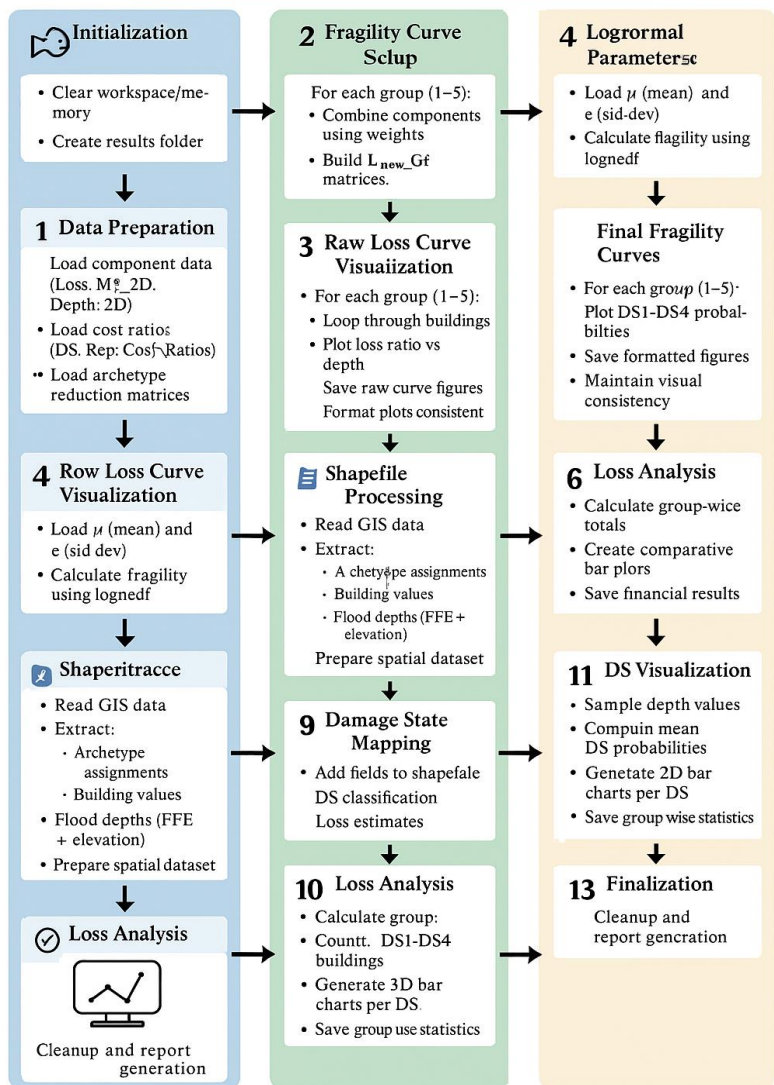
By Jerry Iannelli | October 5, 2017



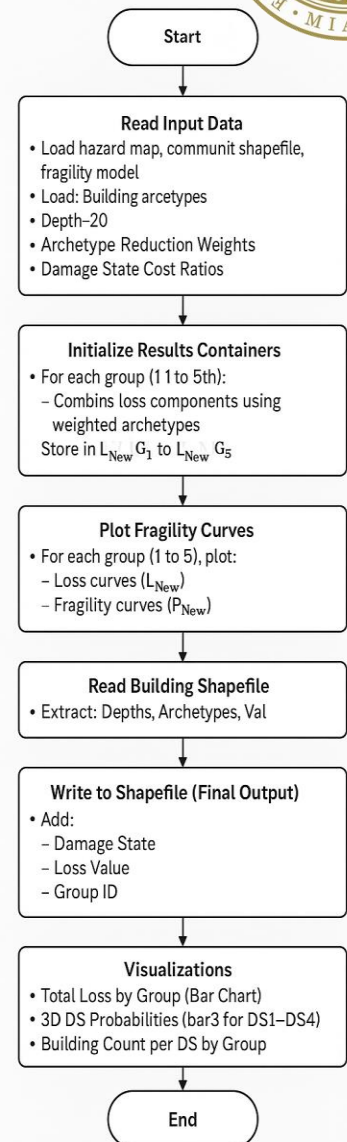
Methodology



Flood Damage Analysis Code Structure



This workflow outlines analyzing building damage by loading hazard and building data, combining loss components for different groups, and plotting fragility curves. The results, including damage states and loss values, are then added to the building shapefile for spatial analysis. Finally, visualizations such as total loss by group and damage state probabilities help summarize the findings.



Archetype Reduction for Flood Damage Analysis

□ Flood Damage Analysis (Archetype Reduction)



□ Group 1 (10 Building Archetypes)

Plot 1: F1 + F2
Plot 2: F3 + F4
Plot 3: F5 + F8
Plot 4: F6 + F7
Plot 5: F9
Plot 6: F10 + F13
Plot 7: F11
Plot 8: F12
Plot 9: F14
Plot 10: F15

□ Group 2 (7 Building Archetypes)

Plot 1: F1 + F2
Plot 2: F3 + F4
Plot 3: F5 + F6
+ F7 + F8
Plot 4: F9 +
F15
Plot 5: F10 +
F13
Plot 6: F11 +
F12
Plot 7: F14

□ Group 3 (4 Building Archetypes)

Plot 1: F1 +
F2 + F3 + F4
Plot 2: F5 +
F6 + F7 + F8
+ F14
Plot 3: F9 +
F10 + F13 +
F15
Plot 4: F11 +
F12

□ Group 4 (3 Building Archetypes)

Plot 1: F1 +
F2 + F3 + F4
Plot 2: F5 +
F6 + F7 + F8
+ F9 + F13 +
F14 + F15
Plot 3: F10 +
F11 + F12

□ Group 5 (1 Building Archetypes)

Plot: F1 + F2
+ F3 + F4 +
F5 + F6 + F7
+ F8 + F9 +
F10 + F11 +
F12 + F13 +
F14 + F15

Our Solution: FloodGuard Miami



FloodGuard Miami is a no-code
flood prediction and alert
platform combining:
Real-time **open data** (NOAA,
USGS, rain gauges)
Community flood reports via
mobile app
AI-driven risk scoring for
neighborhoods

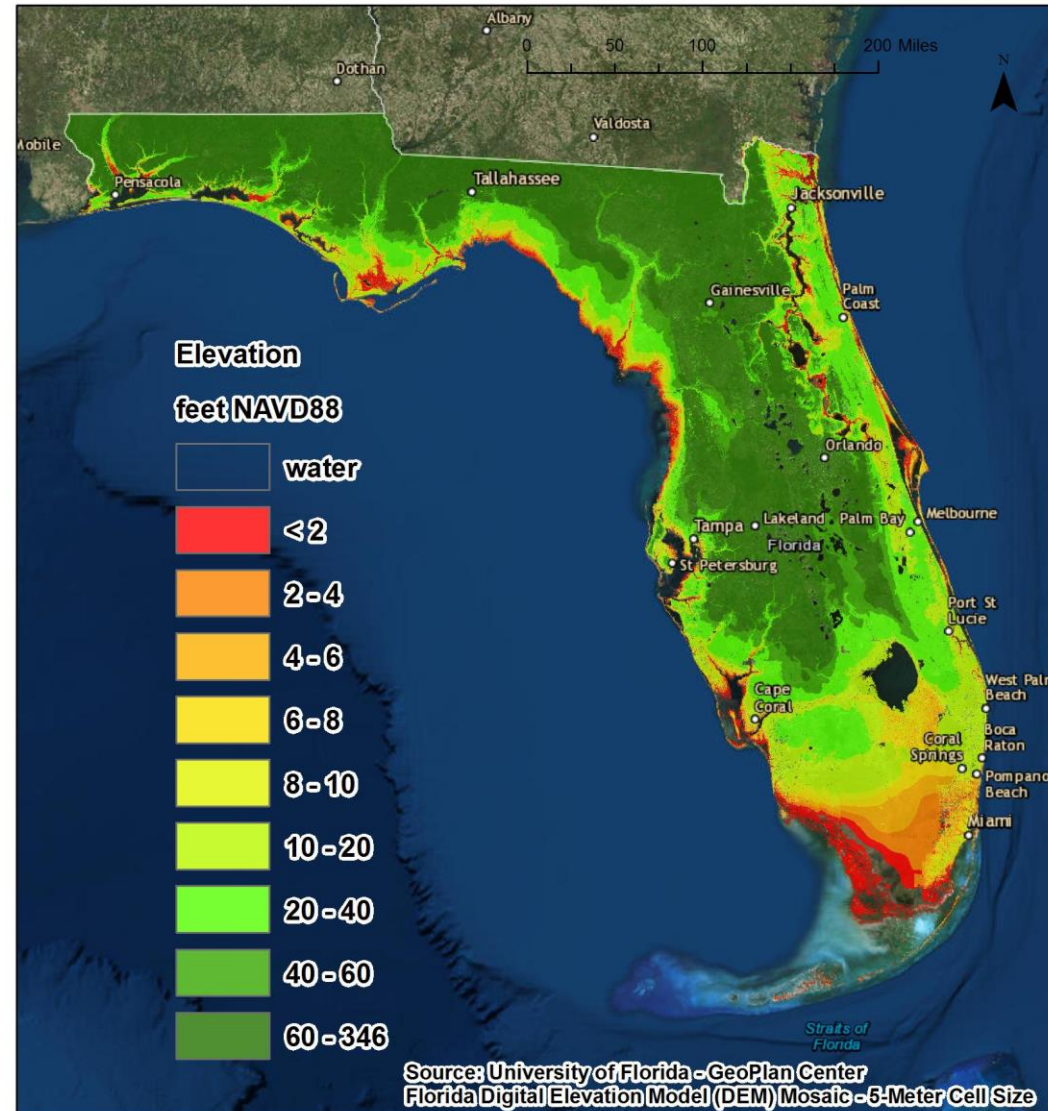
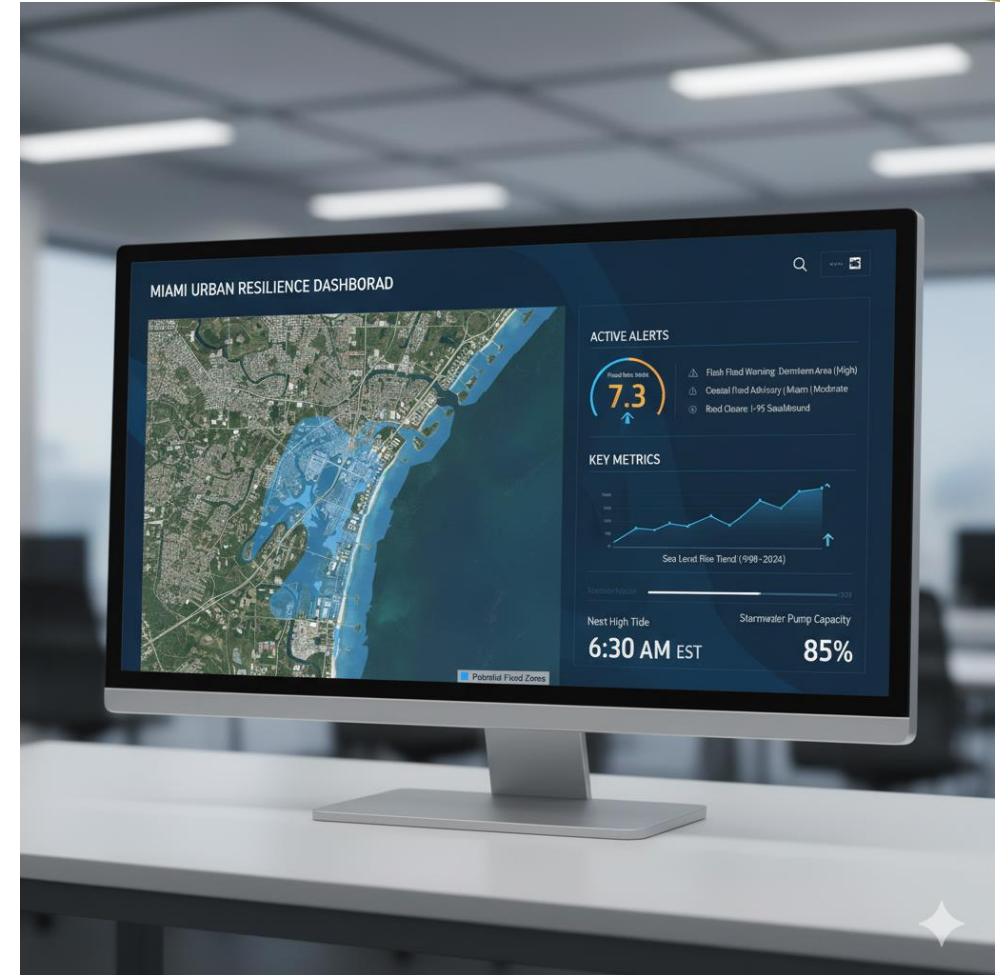


Image Taken: <https://climatecenter.fsu.edu/topics/sea-level-rise>

How It Works



- Collects rainfall & tide data from open APIs
- Integrates local flood reports from users
 - Predicts next 24–48 hr flood risks via simple no-code AI workflows
- Displays alerts + visualization on a public map



Damage States (DS1 to DS4)



<i>DS Level</i>	<i>Functionality</i>	<i>Description</i>	<i>Damage Scale</i>	<i>Damage Ratio</i>
DS1	Limited Occupancy	<i>Water touches the floor joists, up to minor water entering the building. Damage to carpets, pads, baseboards, and flooring. Complete damage to the garage interior is below the first-floor elevation. No major mold or sewer backup.</i>	<i>Slight</i>	<i>0.03 – 0.15</i>
DS2	Restricted Occupancy	<i>Partial damage to drywalls, electrical components (base outlets), water heater, furnace. Complete damage to major equipment, appliances, and furniture on the first floor. Significant damage to the lower bathroom and kitchen cabinets. Potential for major sewer backup and mold issues.</i>	<i>Moderate</i>	<i>0.15 – 0.5</i>
DS3	Restricted Use	<i>Major damage to the non-structural components and interiors within the whole building. Includes drywall damage to upper stories and destruction of electrical switches, mid outlets, bathroom, and kitchen components. Major sewer backup with significant mold issues.</i>	<i>Extensive</i>	<i>0.5 – 0.7</i>
DS4	Restricted Entry	<i>Significant structural damage, including studs, trusses, and joists. Non-structural components destroyed, including drywall, appliances, cabinets, furniture, and more. Damage to rooftop units. The foundation could be compromised and may need to be demolished or replaced.</i>	<i>Complete</i>	<i>0.7 – 1.0</i>

Market & Audience



❑ Segment

❑ City Governments

❑ Real Estate / Insurance

❑ Residents / NGOs

❑ Value

❑ \$10M+ in flood mapping contracts

❑ \$1.5B+ market

❑ Awareness + safety

❑ Target

❑ Miami-Dade, Fort Lauderdale

❑ Developers & agents

❑ General users



Business Model



☐ Tier	☐ Description	☐ Pricing
☐ Free Tier	☐ Public flood alerts	☐ \$0
☐ Pro Tier	☐ City dashboard + predictive insights	☐ \$499/month
☐ Enterprise Tier	☐ Insurance + property API	☐ Custom pricing



Go-to-Market Plan



Phase 1: Pilot with FIU Civil Engineering Dept. + test in Little River

Phase 2: Partner with Miami-Dade Office of Resilience

Phase 3: Expand to Broward & Palm Beach counties

This phase focuses on research and development, with a specific pilot test in the Little River area of Miami.

- University-led research:** FIU's Civil Engineering department would lead the development and testing of solutions for climate-ready infrastructure. FIU has existing research institutes, like the Institute for Resilient and Sustainable Coastal Infrastructure (InteRaCt), that are designed to find engineering solutions for aging and vulnerable infrastructure.
- Testing site:** The Little River Adaptation Action Area, located in Miami-Dade County, has been a key site for planning and pilot projects aimed at localized sea level rise strategies. Initiatives here include investigating septic-to-sewer conversion and creating "blue and green spaces" for water retention.

After a successful pilot, the project scales up through a formal partnership with the Miami-Dade Office of Resilience.

- Expansion of scope:** Miami-Dade County's Office of Resilience has a wide-ranging program to address climate change impacts like sea-level rise and extreme heat, and it already collaborates with FIU. This partnership would allow the solutions developed by FIU to be integrated into larger, county-wide resilience strategies.
- Shared resources:** The collaboration would leverage the resources of both organizations. The Tech Hub project, which FIU is part of, has received millions in grant funding from the U.S. Department of Commerce to support resilient infrastructure.

In the final phase, the initiative is expanded to cover the broader South Florida region.

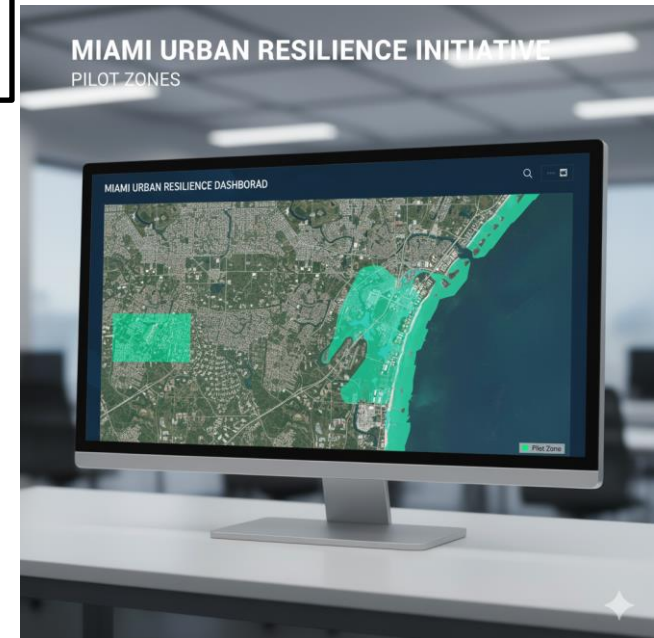
- Regional collaboration:** The expansion would include Broward and Palm Beach counties, building on existing partnerships. These counties are part of the larger regional **Risk and Resilience Tech Hub** alongside Miami-Dade, and the program is designed to scale solutions across the entire region.
- Large-scale implementation:** By expanding, the project would transition from a localized pilot to a regional implementation effort. This would allow the tested technologies and strategies to be commercialized and scaled to benefit a much larger population.

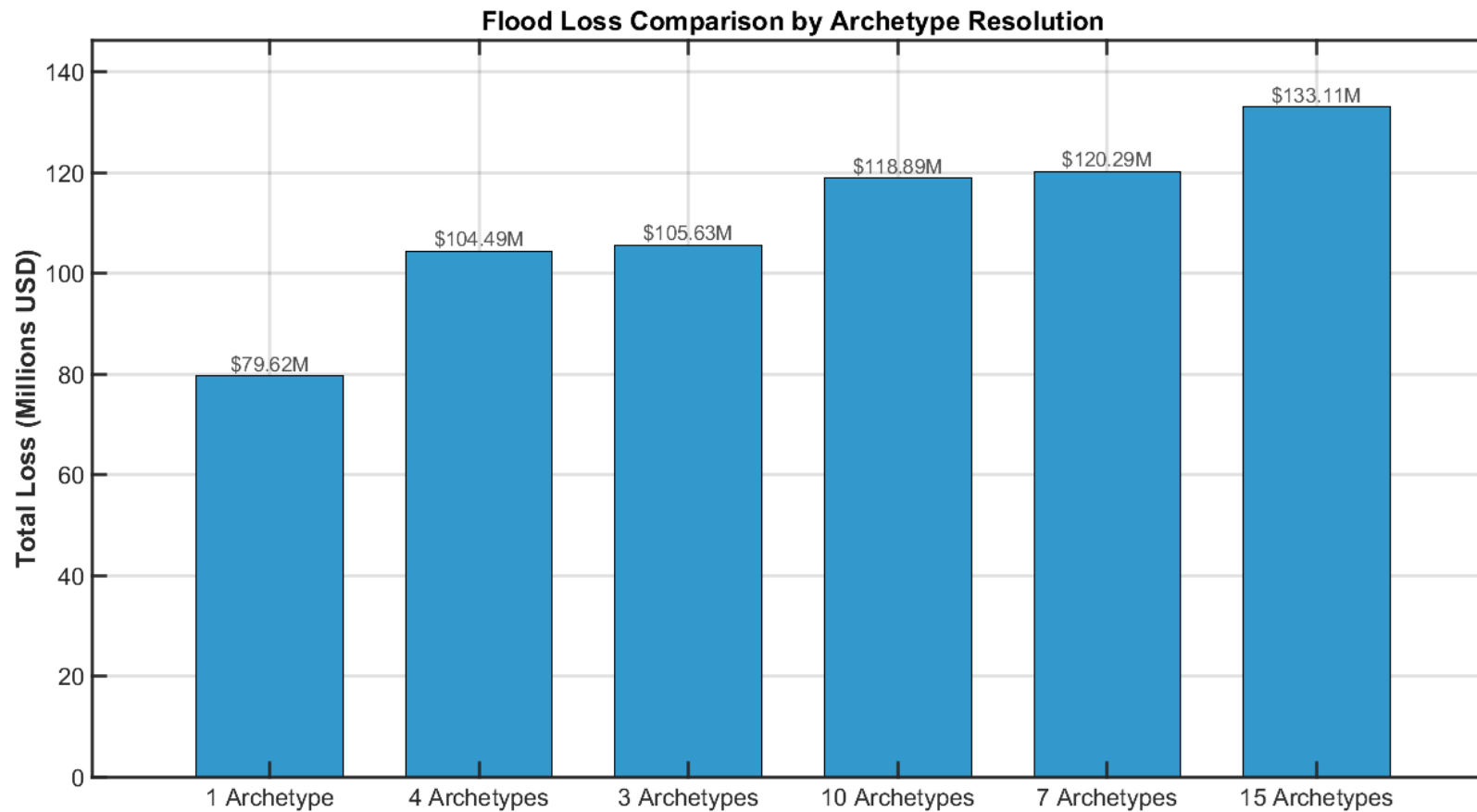
Competitive Advantage



- ✓ Localized data model — built *for* Miami
- ✓ Community-powered reports — real-time accuracy
- ✓ No-code & scalable — fast deployment
- ✓ Civil-engineering expertise — validated design

FEMA's disaster management tools are designed for the specific federal, state, and local government structure of the United States. Global tools, such as those used by the United Nations Office for the Coordination of Humanitarian Affairs (OCHA), are built to handle international coordination across diverse governmental and non-governmental actors.





This chart compares flood loss estimates using different numbers of archetypes in the model. The highest loss is observed with 15 archetypes, showing the importance of detailed categorization for accurate loss assessment.

- Reduce annual property losses by **10–20%**
 - Protect 2.7M Miami residents
- Support Miami's **2050 climate adaptation goals**
 - Bridge citizens, engineers, and policymakers



1. **NOAA National Weather Service (NWS).** (2024). *Flood Safety and Data Resources*. Retrieved from <https://www.weather.gov/safety/flood>
2. **City of Miami Office of Resilience.** (2023). *Miami Forever Climate Ready Strategy*. Retrieved from <https://www.miamigov.com/ClimateReady>
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5. **Florida International University (FIU) Extreme Events Institute.** (2023). *Flood and Hurricane Risk Research*. Retrieved from <https://eei.fiu.edu/>



About the Founder **Mostafa Sameer**

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Florida International University (FIU)

[https://sites.google.com/site/mostafasameeraustpro
file/topics-of-interests-research-based](https://sites.google.com/site/mostafasameeraustprofile/topics-of-interests-research-based)

EmpowHer Mini HackDAY 2025 – In-Person at Florida International University - FIU (Oct 18)

FLORIDA INTERNATIONAL UNIVERSITY, 11200 SW 8th St, Miami, FL 33199, MIAMI, 33199



Mostafa is a civil engineer and researcher specializing in flood risk mitigation, natural hazards, and resilient infrastructure. At FIU, he works on innovative approaches to urban resilience and community-based adaptation strategies for coastal cities like Miami. He founded **FloodGuard Miami** to bridge the gap between scientific data and real-world preparedness — empowering residents and city planners to predict, prepare for, and prevent flood damage through data-driven solutions.

Organizers



BRIAN BAZURTO

Organizer | President, GDG at FIU

[See bio](#)



Amilcar Pena

Team Member

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Bryan A Garcia

Team Member

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Ruben Otano

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